

Comparing SDLC Approaches for the “HablaMundo” Project

Step 1: Analyze the Project Scenario and Identify Its Key Characteristics

Approach

Agile and SAP Activate are the most suitable approaches for the HablaMundo project. Agile provides flexibility for developing AI-driven language learning features and allows iterative improvements based on user feedback. SAP Activate is particularly helpful for integrating new functionality into an existing enterprise system because it provides structured guidance, best practices, and tools that support integration. In contrast, the Waterfall approach may be too rigid for a project that requires continuous feedback and adaptation.

Requirements

The HablaMundo feature must integrate seamlessly with the existing employee learning management system. The platform should use existing employee information such as job roles and geographic locations to personalize language learning. Additionally, the feature must support accessibility for employees with disabilities and include AI-based functionality delivered through cloud-based tools.

Planning

The planning stage should focus on analyzing the architecture of the existing learning system and identifying integration points. The team must determine how the new language learning feature will interact with current employee data and services. Planning should also include selecting appropriate cloud-based AI tools and ensuring compliance with accessibility standards.

Development

Development will involve creating the language learning functionality and integrating it into the current learning platform. The system must securely use employee data to personalize content. AI components should be implemented using cloud-based services to enhance language learning experiences.

Testing

Testing will be a critical stage because the feature must work reliably within the existing system. The team should conduct integration testing, functional testing, accessibility testing, and performance testing. AI components should also be validated to ensure they operate correctly and provide useful learning recommendations.

Customer Involvement

Several stakeholder groups will be involved in the project, including HR managers, system administrators, employees using the learning platform, and accessibility experts. Their feedback is essential to ensure the system meets organizational needs and remains accessible for all employees.

Change Management

Since the project integrates with an existing system, changes must be carefully managed. Updates to AI functionality, accessibility improvements, and integration modifications must be implemented while ensuring the existing learning platform continues to operate without disruption.

Risk Management

Potential risks include integration issues between the new feature and the existing system, performance issues related to AI functionality, accessibility compliance challenges, and security risks associated with employee data.

Documentation

Comprehensive documentation should be created for the integration architecture, AI functionality, accessibility compliance, and configuration processes. Proper documentation will ensure that the system can be maintained and updated efficiently in the future.

Suitable Projects

The HablaMundo project is best suited for methodologies that support integration with existing systems, continuous feedback, and the ability to adapt to evolving requirements.

Project Speed

The project requires a balanced development speed. Rapid iterations can help deliver features quickly, but quality, integration stability, and accessibility must remain top priorities.

Focus

The main focus of the project is to successfully integrate an AI-powered language learning feature into an existing enterprise learning platform while ensuring accessibility, maintainability, and a seamless user experience.

Step 2: Compare the Three SDLC Approaches

Step 2.1 Traditional (Waterfall)

Strengths for HablaMundo

The Waterfall approach provides clear structure and detailed planning. This can help define integration requirements and technical specifications early in the project. Strong documentation may also support coordination among teams working on a complex enterprise system.

Weaknesses for HablaMundo

Waterfall lacks flexibility when requirements change. Since the HablaMundo project involves AI functionality and accessibility considerations that may evolve over time, this rigid structure could limit adaptability. Additionally, integration challenges with the existing system may only become visible later in the development process.

Step 2.2 Agile

Strengths for HablaMundo

Agile supports iterative development and continuous feedback from stakeholders. This approach allows the team to test and improve the language learning feature in smaller increments. Agile also helps incorporate feedback from employees and accessibility experts more effectively.

Weaknesses for HablaMundo

Agile typically produces less formal documentation compared to other methodologies. For enterprise environments that require strong documentation and system integration guidelines, this could create challenges. In addition, managing frequent stakeholder involvement across a global organization may require significant coordination.

Step 2.3 SAP Activate

Strengths for HablaMundo

SAP Activate combines structured planning with iterative implementation. It provides best practices, guided configuration, and integration support designed for enterprise systems. This approach can help ensure that the new feature integrates smoothly with the existing learning platform while maintaining system stability.

Weaknesses for HablaMundo

SAP Activate is often designed for full enterprise implementations, so it may be somewhat specialized for projects focused mainly on adding a single feature. Additionally, its emphasis on standard solutions may limit flexibility for innovative AI features.

Step 3: Choose the Most Appropriate Approach

The most suitable approach for the HablaMundo project is SAP Activate combined with Agile principles. The project involves integrating a new AI-based language learning feature into an existing enterprise learning system, which requires both structured planning and flexibility. SAP Activate provides best practices and guided integration tools that help ensure compatibility with the current system. At the same time, Agile principles allow the development team to work iteratively, gather feedback from employees and accessibility experts, and continuously improve the language learning functionality. This hybrid approach balances stability and adaptability, which is critical for enterprise environments. Waterfall would be less suitable because its rigid structure makes it difficult to adjust to evolving AI capabilities and user needs. Therefore, combining SAP Activate with Agile practices offers the best balance between integration reliability and development flexibility.
